

1. Problem Tackled

This machine is a **computational** Turing Machine. Its job is to read a string of **a's** and **b's** on Tape 1 and produce a translated version on Tape 2, converting **a**→**c** and **b**→**d**. It halts in the accept state once the whole input has been processed.

2. Reference for the Problem

There wasn't any specific reference for this problem, but I took inspiration from questions I did in the past from Theory of Computation, 3rd edition by Michael Sipser

3. How the Machine Works (In Words)

The machine uses **two tapes**:

- **Tape 1** holds the input. The head moves right through the entire string.
- **Tape 2** starts blank and stores the output.

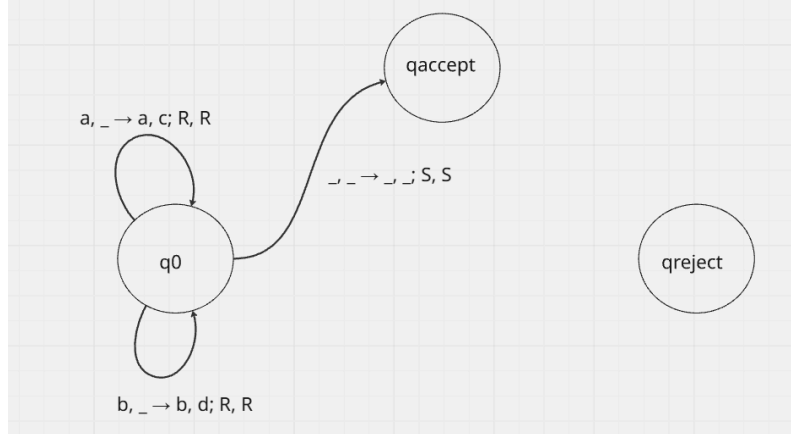
For each symbol on Tape 1:

- If it reads **a**, it writes **c** on Tape 2.
- If it reads **b**, it writes **d** on Tape 2.

Both heads then move right. When Tape 1 reaches , the machine enters the accept state. If the machine sees any symbol outside {a, b}, it prints an error.

4. State Diagram

2 Tape Turing Machine that replaces a with c and b with d



5. Verifying Correct Operation

To verify correct operation, I ran the machine on four different inputs and checked both tapes after execution. For **aabb**, the machine read each character, produced **ccdd** on Tape 2, and accepted. For **aaaaab**, it translated the five a's into **ccccc** and the final b into **d**, producing **ccccc d** before accepting. For **cba**, the machine immediately flagged **c** as an illegal symbol, printed an error, and halted without producing output on Tape 2, which remained blank. For **bbba**, it correctly translated the three b's to **ddd** and the final a to **c**, producing **ddd c** and accepting.